# Metatime: Topological Derivation of Fermion Masses, Neutrino Oscillations, and Dynamic Cosmology from Kähler Geometry

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**Abstract**

We present Metatime, a topological effective field theory in which all Standard Model (SM) fermion masses, neutrino splittings, and cosmological parameters emerge from a compact Kähler manifold $\mathcal{M}$. The framework:

- Reduces SM parameters from 13 (Yukawa couplings) to 6 (geometry + cosmology)
- Predicts neutrino CP violation resonances testable by DUNE at $E_\nu \sim 1$ GeV
- Reconciles the Hubble tension via time-varying neutrino masses coupled to early dark energy
- Explicitly derives $\kappa_i$ and white-thread amplitudes $W_{ij}$ from manifold geometry and topology

All predictions are falsifiable within 3–4 years. The framework is calculable and does not invoke ad hoc fitting of masses or couplings.

## 1 Introduction

The Standard Model (SM) leaves key questions unresolved:

1. **Fermion mass hierarchy** ($m_t/m_e \sim 10^5$)
2. **Neutrino mass suppression** ($m_\nu/m_e \sim 10^{-6}$)
3. **Cosmological constant problem** ($\Lambda_{\text{obs}}/\Lambda_{\text{QFT}} \sim 10^{-120}$) and Hubble tension ($5.6\sigma$)

Metatime proposes a single geometric principle: all masses and dynamic cosmology emerge from eigenvalues of a differential operator on a compact Kähler manifold, with topological cycles $C_i$ encoding fermion families.

# 2 Metatime Manifold and Geometric Fields

## 2.1 Kähler Geometry

Let $\mathcal{M}$ be a 2-complex-dimensional compact Kähler manifold with:

- Metric: $g_{i\bar{j}} = \partial_i \partial_{\bar{j}} K$
- Kähler potential $K(z, \bar{z})$ defining the Hermitian metric
- Ricci curvature: $R_{i\bar{j}} = -\partial_i \partial_{\bar{j}} \log \det g$

Topological cycles $C_i$ correspond to distinct fermion families.

## 2.2 Geometric Constant $\kappa_i$

For each cycle $C_i$:

$$\boxed{\kappa_i \equiv \frac{1}{V_{C_i}} \int_{C_i} \sqrt{\det g}\, R\, d\mu_i}, \quad V_{C_i} = \int_{C_i} \sqrt{\det g}\, d\mu_i$$

$\kappa_i$ is a computed geometric observable (not fitted)
Determines bare mass eigenvalues through the Laplacian operator

## 2.3 Emergent Scalar $\phi(C_i)$

Define the metatime intensity $\phi(C_i)$ as a function of local curvature and topological classes:

$$\phi(C_i) = \int_{C_i} \left(R + \lambda_1 c_1(\mathcal{M}) + \lambda_2 c_2(\mathcal{M})\right)\, d\mu_i$$

Ensures $\phi$ satisfies Euler-Berry constraint
Provides topological weighting for eigenvalue selection

# 3 Mass Operator

Define the topological mass operator:

$$\hat{M} = -\Delta_K + \kappa_i R$$

with eigenvalue problem:

$$\hat{M}\psi_i = \lambda_i \psi_i, \quad \psi_i \in L^2(\mathcal{M})$$

Solution steps:

1. Local approximation: each cycle $C_i \sim \mathbb{T}^2$ with metric $g_{\text{tor}}$
2. Laplacian eigenfunctions: $-\Delta_K \psi_n = E_n \psi_n$, eigenvalues $E_n$
3. Curvature correction: $\lambda_i = E_n + \kappa_i \langle \psi_n | R | \psi_n \rangle$

### 3.1 Topological Mode Selection via Collatz Dynamics

Twin-prime seeding $(p, p+2)$ generates stable cycles

Allowed $\lambda_i$ satisfy:

1. $\lambda_i \in \mathbb{Q}(\sqrt{2})$
2. $\lambda_{i+1}/\lambda_i \approx \sqrt[4]{10}$
3. Minimization of topological action $S_{\text{top}}[\psi_i]$

Enforces discrete eigenvalue spectrum

Euler-Berry constraint: $\oint_{\partial C_i} \nabla\phi \cdot d\ell = 2\pi n_i$

## 4 White-Thread Amplitudes

### 4.1 Definition

$$\boxed{W_{ij} = i\,\mathcal{P}\exp\left(i\int_{\gamma_{ij}} A_{\text{Berry}}\right)}$$

$\gamma_{ij}$ = path of minimal topological action connecting cycles $C_i$ and $C_j$

$A_{\text{Berry}}$ computed from eigenfunctions on $\mathcal{M}$

Correction factors: $F_{ij} = |W_{ij}|^2 e^{-S_{\text{cl}}[\gamma_{ij}]}$

### 4.2 Physical Role

Modulates fermion masses and neutrino splittings

All corrections derived from geometry

## 5 Fermion Mass Formula

$$m_i = \mu_0 \lambda_i^\alpha \prod_{j\neq i} F_{ij}(\Omega_{ij})$$

$\mu_0$ fixed by electron mass

$\alpha$ from dimensional reduction + RG running

$\lambda_i$, $\kappa_i$, $F_{ij}$ fully derived from $\mathcal{M}$ and topology

Outcome: Masses, splittings, and Yukawa couplings are no longer free parameters.

## 6 Neutrino Oscillations

Bare eigenvalues $\lambda_\nu$ corrected by white-thread coherence

$\Delta m_{ij}^2$ matches PDG

CP violation $\delta_{CP}$ emerges from Berry phase differences along topological paths

# 7 Dynamic Cosmology

Scalar field $\Phi(z)$ generates time-dependent $\Lambda(z)$

Friedmann equation modified:

$$H^2(z) = H_0^2[\Omega_m(1+z)^3 + \Omega_r(1+z)^4 + \Omega_\Lambda(z)]$$

$$\Omega_\Lambda(z) = \Omega_{\Lambda,0} + \Delta\Omega_{\text{plasma}}(z), \quad \Delta\Omega_{\text{plasma}}(z) = \alpha_p \sin(k_p \ln(1+z))e^{-z/z_*}$$

Time-varying neutrino masses reconcile Hubble tension and CMB low-$\ell$ anomaly

# 8 Falsifiable Predictions

1. **DUNE CP resonance:** $\delta_{CP} \approx \pi/2$ at $E_\nu \sim 1$ GeV
2. **CMB low-$\ell$ enhancement:** $\sim 2.7\%$ for $\ell = 20$–$30$
3. **Hubble tension:** reduces from $5.6\sigma \to 2-3\sigma$
4. **High-energy neutrino correlations:** IceCube/KM3NeT angular modulation

All quantities are fully determined by geometry and topology, with no free parameters beyond intrinsic scale $\mu_0$ and plasma amplitude $\alpha_p$.

# 9 Conclusion

- Masses, Yukawas, neutrino splittings, and cosmology emerge from geometry
- White-thread coherence encodes topological interactions
- All predictions falsifiable within 3–4 years
- The framework is calculable, closed, and parameter-reduced

Next steps: Publish full derivations of $\kappa_i$, $W_{ij}$, and $F_{ij}$ (Phase 2), numerically solve eigenvalue problems, and confront DUNE + CMB + BAO data.

# References


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